
Label-Only Membership Inference Attacks

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Abstract

Membership inference is one of the simplest privacy threats faced by machine learning models that are trained on private sensitive data. In this attack, an adversary infers whether a particular point was used to train the model, or not, by observing the model’s predictions. **Whereas current attack methods all require access to the model’s predicted confidence score**, we introduce a label-only attack that instead evaluates the robustness of the model’s predicted (hard) labels under perturbations of the input, to infer membership. Our label-only attack is not only as-effective as attacks requiring access to confidence scores, it also demonstrates that a class of defenses against membership inference, which we call “confidence masking” because they obfuscate the confidence scores to thwart attacks, are insufficient to prevent the leakage of private information. Our experiments show that training with differential privacy or strong ℓ_2 regularization are the only current defenses that meaningfully decrease leakage of private information, even for points that are outliers of the training distribution.

1. Introduction

Machine learning algorithms are often trained on sensitive or private user information, e.g., medical records (Stanfill et al., 2010), conversations (Devlin et al., 2018), or financial information (Ngai et al., 2011). Trained models can inadvertently leak information about their training data (Shokri et al., 2016; Carlini et al., 2019)—violating users’ privacy.

In perhaps the simplest form of information leakage, *membership inference (MI)* (Shokri et al., 2016) attacks enable an adversary to determine whether or not a data point was used in the training data. Revealing just this information can

cause harm—it leaks information about specific individuals instead of the entire population. Consider a model trained to learn the link between a cancer patient’s morphological data and their reaction to some drug. An adversary with a victim’s morphological data and query access to the trained model cannot directly infer if the victim has cancer. However, inferring that the victim’s data was part of the model’s training set reveals that the victim indeed has cancer.

Existing MI attacks exploit the higher confidence that models exhibit on their training data (Pyrgelis et al., 2017; Truex et al., 2018; Hayes et al., 2019; Salem et al., 2018). An adversary queries the model on a candidate data point to obtain the model’s confidence and infers the candidate’s membership in the training set based on a decision rule. The difference in prediction confidence is largely attributed to *overfitting* (Shokri et al., 2016; Yeom et al., 2018).

A large body of work has been devoted to understanding and mitigating MI leakage in ML models. Existing defense strategies fall into two broad categories and either

- (1) reduce overfitting (Truex et al., 2018; Shokri et al., 2016; Salem et al., 2018); or,
- (2) perturb a model’s predictions so as to minimize the success of known membership attacks (Nasr et al., 2018a; Jia et al., 2019; Yang et al., 2020).

Defenses in (1) use regularization techniques or increase the amount of training data to reduce overfitting. In contrast, the adversary-aware defenses of (2) explicitly aim to minimize the MI advantage of a particular attack. They do so either by modifying the training procedure (e.g., an additional loss penalty) or the inference procedure after training. These defenses implicitly or explicitly rely on a strategy that we call *confidence-masking*¹, where the MI signal in the model’s confidence scores is masked to thwart existing attacks.

We introduce *label-only MI attacks*. Our attacks are more general: an adversary need only obtain (hard) labels—without prediction confidences—of the trained model. This threat model is more realistic, as ML models deployed in user-facing products need not expose raw confidence scores. Thus, our attacks can be mounted on any ML classifier.

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¹Similar to *gradient masking* from the adversarial examples literature (Papernot et al., 2017).

In the label-only setting, a naive baseline predicts misclassified points as non-members. Our focus is surpassing this baseline. To this end, we will have to make *multiple* queries to the target model. We show how to extract fine-grained MI signal by analyzing a model’s robustness to perturbations of the target data, which reveals signatures of its decision boundary geometry. Our adversary queries the model for predicted labels on augmentations of data points (e.g., translations in vision domains) as well as adversarial examples.

We make the following contributions. In § 5.1, we introduce the first label-only attacks, which match confidence-vector attacks. By combining them, we outperform all others. In § 5.2, 5.3 and 5.4, we show that *confidence masking* is not a viable defense to privacy leakage, by breaking two canonical defenses that use it—MemGuard and Adversarial Regularization. In § 6, we evaluate two additional techniques to reducing overfitting: data augmentation and transfer learning. We find that data augmentation can worsen MI leakage while transfer learning can mitigate it. In § 7, we introduce “outlier MI”: a stronger property that defenses should satisfy to protect MI of worst-case inputs; at present, differentially private training and (strong) L2 regularization appear to be the only effective defenses. Our code is available at <https://github.com/cchoquette/membership-inference>.

2. Background and Related Works

Membership inference attacks (Shokri et al., 2016) are a form of privacy leakage that identify if a given data sample was in a machine learning model’s training dataset. Given a sample x and access to a trained model h , the adversary uses a classifier or decision rule f_h to compute a membership prediction $f(x; h) \in \{0, 1\}$, with the goal that $f(x; h) = 1$ whenever x is a training point. The main challenge in mounting a MI attack is creating the attack classifier f , under various assumptions about the adversary’s knowledge of h and its training data distribution. Most prior work assumes that an adversary has only black-box access to the trained model h , via a query interface that on input x returns part or all of the confidence vector $h(x) \in [0, 1]^C$ (for a classification task with C classes).

The attack classifier f is often trained on a local **shadow** (or, source) model $\hat{h}_i$, which is trained on the same (or a similar) distribution as h ’s training data. Because the adversary trained $\hat{h}_i$, they can assign membership labels to any input x , and use this dataset to train f . Salem et al. (2018) later showed that this strategy succeeds even when the adversary only has data from a different, but similar, task and that **shadow** models are unnecessary: a threshold predicting $f(x; h) = 1$ when the max prediction confidence, $\max_i h(x)_i$, is above a tuned threshold, suffices.

Yeom et al. (2018) investigate how querying related inputs x' to x can improve MI. Song et al. (2019) explore how models explicitly trained to be robust to adversarial examples can become more vulnerable to MI (similar to our analysis of data augmentation in § 6). Both works are crucially different because they use a different attack methodology and assume access to the confidence scores. Sablayrolles et al. (2019) demonstrate that black-box attacks (like ours) can approximate white-box attacks by effectively estimating the model loss for a data point. Refer to Appendix § A for a detailed background, including on defenses.

3. Attack Model Design

Our proposed MI attacks improve on existing attacks by (1) combining *multiple strategically perturbed samples* (queries) as a fine-grained signal of the model’s decision boundary, and (2) operating in a label-only regime. Thus, our attacks pose a threat to *any* query-able ML service.

3.1. A Naive Baseline: The Gap Attack

Label-only MI attacks face a challenge of granularity. For any query x , our attack model’s information is limited to only the predicted class-label, $\arg\max_i h(x)_i$. A simple *baseline* attack (Yeom et al., 2018)—that predicts any misclassified data point as a non-member of the training set—is a useful benchmark to assess the extra (non-trivial) information that MI attacks, label-only or otherwise, can extract. We call this baseline the *gap attack* because its accuracy is directly related to the gap between the model’s accuracy on training data ($\text{acc}_{\text{train}}$) and held out data (acc_{test}):

$$1/2 + (\text{acc}_{\text{train}} - \text{acc}_{\text{test}})/2, \quad (1)$$

where $\text{acc}_{\text{train}}, \text{acc}_{\text{test}} \in [0, 1]$. To exploit additional leakage on top of this baseline attack (achieve non-trivial MI), any label-only adversary must necessarily make additional queries to the model. To the best of our knowledge, this trivial baseline is the only attack proposed in prior work that uses only the predicted label, $y = \arg\max_i h(x)_i$.

3.2. Attack Intuition

Our strategy is to compute label-only “proxies” for the model’s confidence by evaluating its *robustness* to strategic input perturbations of x , either synthetic (i.e., data augmentation) or adversarial (examples) (Szegedy et al., 2013). Following a max-margin perspective, we predict that data points that exhibit high robustness are training data points. Works in the adversarial example literature share a similar perspective that non-training points are closer to the decision boundary and thus more susceptible to perturbations (Tanay & Griffin, 2016; Tian et al., 2018; Hu et al., 2019).

Our intuition for leveraging robustness is two-fold. First,

models *trained with* data augmentation have the capacity to overfit to them (Zhang et al., 2016). Thus, we evaluate any “effective” train-test gap on the augmented dataset by evaluating x and its augmentations, giving us a more fine-grained MI signal. For models not trained using augmentation, their robustness to perturbations can be a proxy for model confidence. Given the special case of (binary) logistic regression models, with a learned weight vector w and bias b , the model will output a confidence score for the positive class of the form: $h(x) := \sigma(w^\top x + b)$, where $\sigma(t) = \frac{1}{1+e^{-t}} \in (0, 1)$ is the logistic function.

Here, there is a monotone relationship between the confidence at x and the Euclidean *distance* to the model’s decision boundary. This distance is $(w^\top x + b)/\|w\|_2 = \sigma^{-1}(h(x))/\|w\|_2$. Thus, obtaining a point’s distance to the boundary yields the same information as the confidence score. Computing this distance is exactly the problem of finding the smallest *adversarial perturbation*, which can be done using label-only access to a classifier (Brendel et al., 2017; Chen et al., 2019). Our thesis is that this relationship will persist for deep, non-linear models. This thesis is supported by prior work that suggests that deep neural networks can be closely approximated by linear functions in the vicinity of the data (Goodfellow et al., 2014).

3.3. Data Augmentation

Our data augmentation attacks create a MI classifier $f(x; h)$ for a model h . Given a target point (x_0, y_{true}) , the adversary trains f to output $f(x_0, h) = 1$, if x_0 was a training member. To do this, they tune f to maximize MI accuracy on a source (or “shadow”) model assuming knowledge of the target model’s architecture and training data distribution. They then “transfer” f to perform MI by querying the black-box model h . Using x_0 , we create additional data points $\{\hat{x}_1, \dots, \hat{x}_N\}$ via different data augmentation strategies, described below. We query the target model h ($\hat{h}$ in tuning) to obtain labels $(y_0, y_1, \dots, y_N) \leftarrow (h(x_0), h(\hat{x}_1), \dots, h(\hat{x}_N))$. Let $b_i \leftarrow \mathbb{I}(y_{\text{true}} = (y_i))$ be the indicator function for whether the i -th queried point was misclassified. Finally, we apply $f(b_0, \dots, b_N) \rightarrow \{0, 1\}$ to classify x_0 .

We experiment with two common data augmentations in the computer vision domain: image rotations and translations. For **rotations**, we generate $N = 3$ images as rotations by a magnitude $\pm r^\circ$ for $r \in [1, 15]$. For **translations**, we generate $N = 4d + 1$ translated images satisfying $|i| + |j| = d$ for a pixel bound d , where we horizontal shift by $\pm i$ and vertical shift by $\pm j$. In both we include the source image.

3.4. Decision Boundary Distance

These attacks predict membership using a point’s distance to the model’s decision boundary. Here, we extend the

intuition that this distance can be a proxy for confidence of linear models (see § 3.2) to deep neural networks.

Recall that confidence-thresholding attacks predict highly confident samples as members (Salem et al., 2018). Given some estimate $\text{dist}_h(x, y)$ of a point’s ℓ_2 -distance to the model’s boundary, we predict x a member if $\text{dist}_h(x, y) > \tau$ for some threshold τ . We define $\text{dist}_h(x, y) = 0$ for misclassified points, where $\arg\max_i h(x)_i \neq y$, because no perturbation was needed. We tune τ on a shadow $\hat{h}_i$, and find that even crude estimates, e.g., Gaussian noise, can lead to nearly comparable attacks (see § 5.5). We now discuss methods for estimating $\text{dist}(x, y)$.

A White-Box Baseline for estimating $\text{dist}(x, y)$ is an idealized *white-box* attack and is therefore not label-only. We use adversarial-examples generated by the Carlini and Wagner attack (Carlini & Wagner, 2017): given (x, y) the attack tries to find the closest point x' to x in the Euclidean norm, such that $\arg\max h(x') \neq y$.

Label-only attacks use only black-box access. We rely on label-only adversarial example attacks (Brendel et al., 2017; Chen et al., 2019). These attacks start from a random point x' that is misclassified, i.e., $h(x') \neq y$. They then “walk” along the boundary while minimizing the distance to x . We use “HopSkipJump” (Chen et al., 2019), which closely approximates stronger white-box attacks.

Robustness to random noise is a simpler approach based on random perturbations. Again, our intuition stems from linear models: a point’s distance to the boundary is directly related to the model’s accuracy when it is perturbed by isotropic Gaussian noise (Ford et al., 2019). We compute a proxy for $d_h(x, y)$ by evaluating the accuracy of h on N points $\hat{x}_i = x + \mathcal{N}(0, \sigma^2 \cdot I)$, where σ is tuned on $\hat{h}$. For binary features we instead use Bernoulli noise: each $x_j \in x$ is flipped with probability p , which is tuned on $\hat{h}$.

Many signals for robustness can be combined to improve the attack performance. We evaluate $d_h(x, y)$ for augmentations of x from § 3.3. We only evaluate this attack where indicated due to its high query cost (see § 5.5).

4. Evaluation Setup

Our evaluation is aimed at understanding how label-only MI attacks compare with prior attacks that rely on access to a richer query interface. To this end, **we use an identical evaluation setup as prior works** (Nasr et al., 2018b; Shokri et al., 2016; Long et al., 2017; Truex et al., 2018; Salem et al., 2018) (see Appendix § B). We answer the following questions in our evaluation, § 5, § 6 and § 7:

1. Can label-only MI attacks match prior attacks that use

the model’s (full) confidence vector?

2. Are defenses against confidence-based MI attacks always effective against label-only attacks?
3. What is the query complexity of label-only attacks?
4. Which defenses prevent both label-only and full confidence-vector attacks?

To evaluate an attack’s success, we pick a balanced set of points from the task distribution, of which half come from the target model’s training set. We measure attack success as overall MI accuracy but find F1 scores to approximately match, with near 100% recall. See Supplement § B.2 for further discussion on this evaluation.

Overall, we stress that our main goal is to show that *in settings where MI attacks have been shown to succeed*, a label-only query interface is sufficient. In general, we should not expect our label-only attacks to *exceed* the performance of prior MI attacks since the former uses strictly less information from queries than the latter. There are three notable exceptions: our combined attack² (§ 5.1), “confidence masking” defenses (§ 5.2), and models trained with significant data augmentation (§ 6.1). In the latter two cases, we find that existing attacks severely underestimate MI.

4.1. Attack Setup

We provide a detailed account of model architectures and training procedures in Supplement § B.1 and of our threat model in Supplement § C. We evaluate our attacks on 8 datasets used by the canonical work of Shokri et al. (2016). These include 3 computer vision tasks³, which are our main focus because of the importance of data augmentation to them, and 4 non-computer-vision tasks⁴ to showcase our attacks’ transferability. We train target neural networks on subsets of the original training data, exactly as performed by Shokri et al. (2016) and several later works (in both data amount and train-test gap). Controlling the training set size lets us control the amount of overfitting, which strongly influences the strength of MI attacks (Yeom et al., 2018). Prior work has almost exclusively studied (confidence-based) MI attacks on these small datasets where models exhibit a high degree of overfitting. Recall that our goal is to show that label-only attacks match confidence-based approaches: scaling MI attacks (whether confidence-based or label-only) to

larger training datasets is an important area of future work.

5. Evaluation of Label-Only Attacks

5.1. Label-Only Attacks Match Confidence-Vector Attacks

We first focus on question 1). Recall from § 3.1 that any label-only attack (with knowledge of y) is always trivially lower-bounded by the baseline gap attack of Yeom et al. (2018), predicting any misclassified point as a non-member.

Our main result is that our label-only attacks consistently outperform the gap attack and perform on-par with prior confidence-vector attacks; by combining attacks, we can even **surpass** the canonical confidence-vector attacks.

Observing Figure 1 and Table 1 (a) and (c), we see that the *confidence-vector attack* outperforms the baseline *gap attack*, demonstrating that it exploits non-trivial MI. Remarkably, we find that our *label-only boundary distance attack* performs at least on-par with the confidence-vector attack. Moreover, our simpler but more query efficient (see § 5.5) *data augmentation attacks* also consistently outperform the baseline but fall short of the confidence-vector attacks. Finally, combining these two label-only attacks, we can consistently outperform every other attack. These models were not trained with data augmentation; in § 6.1, we find that when they are, our data augmentation attacks outperform all others. Finally, we verify that as the training set size increases, all attacks monotonically decrease because the train-test gap is reduced. Note that on CIFAR-100, we experiment with the largest training subset possible: 30,000 data points, since we use the other half as the source model training set (and target model non-members).

Beyond Images We show that our label-only attacks can be applied outside of the image domain in Table 2. Our label-only attack evaluates a model’s accuracy under *random perturbations*, by adding Gaussian noise for continuous-featured inputs, and flipping binary values according to Bernoulli noise (see § 3.4). Using 10,000 queries, our attacks closely match (at most 4 percentage-point degradation) confidence-based attacks. Note that our attacks could also be instantiated in audio or natural language domains, using existing adversarial examples attacks (Carlini & Wagner, 2018) and data augmentations (Zhang et al., 2015).

5.2. Breaking Confidence Masking Defenses

Answering question 2), we showcase an example where our label-only attacks *outperform* prior attacks by a significant margin, despite the strictly more restricted query interface that they assume. We evaluate *defenses* against MI attacks and show that while these defenses do protect against existing confidence-vector attacks, they have little

²Note that this attack’s performance exceeds prior confidence-vector attacks, but that we do not test its confidence-vector analog. Our results indicate that it should perform comparably.

³MNIST, CIFAR-10, and CIFAR-100: https://www.tensorflow.org/api_docs/python/tf/keras/datasets

⁴Adult Dataset: <http://archive.ics.uci.edu/ml/datasets/Adult>
Texas-100, Purchase-100, and Locations datasets: <https://github.com/privacytrustlab/datasets>

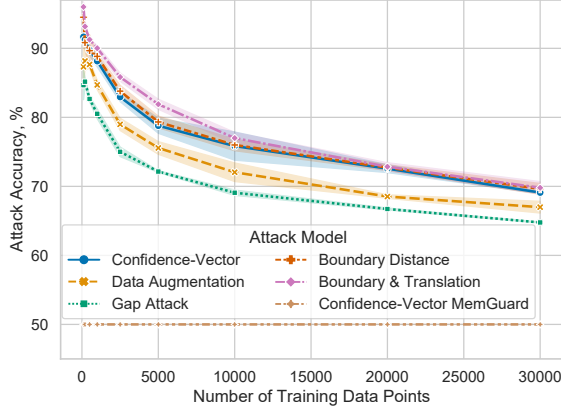


Figure 1. Accuracy of MI attacks on CIFAR-10. We evaluate 100 to 10,000 training points and compare the baseline gap attack, the confidence-vector attack that relies on a fine-grained query interface, and our label-only attacks based on data augmentation and distance to the decision boundary. We also show the confidence-vector attack performance against MemGuard, noting that our label-only performances remain nearly unaltered. For the data augmentation attack, we report the best accuracy across multiple values of r (rotation angle) and d (number of translated pixels).

to no effect on our label-only attacks. Because any ML classifier providing confidences also provides the predicted labels, our attacks fall within their threat model, refuting these defenses’ security claims.

We identify a common pattern to these defenses that we call *confidence masking*, wherein defenses aim to prevent MI by directly minimizing the privacy leakage in a model’s confidence scores. To this end, confidence-masking defenses explicitly or implicitly mask (or, obfuscate) the information contained in the model’s confidences, (e.g., by adding noise) to thwart existing attacks. These defenses, however, *have a minimal effect on the model’s predicted labels*. MemGuard (Jia et al., 2019) and prediction purification (Yang et al., 2020) explicitly maintain the invariant that the model’s predicted labels are not affected by the defense, i.e.,

$$\forall x, \quad \operatorname{argmax} h(x) = \operatorname{argmax} h^{\text{defense}}(x),$$

where h^{defense} is the defended version of the model h .

An immediate issue with the design of confidence-masking defenses is that, by construction, they will not prevent any label-only attack. Yet, these defenses were reported to drive the success rates of existing MI attacks to within chance. This result suggests that prior attacks fail to properly extract membership information contained in the model’s predicted labels, and implicitly contained within its scores. Our label-only attack performances clearly indicate that confidence masking is not a viable defense strategy against MI.

We show that two peer-reviewed defenses, MemGuard (Jia et al., 2019) and adversarial regularization (Nasr et al.,

Table 1. Accuracy of MI attacks on CIFAR-100 and MNIST. The target models are trained using 30,000 data points for CIFAR-100 and 1,000 for MNIST. Tables (a) and (c) report results without any defense; (b) and (d) with MemGuard (Jia et al., 2019), which prevents the confidence-vector attacks via “confidence masking”. ‘Combined’ refers to the boundary and translation attack. Results that are affected by confidence masking are marked in red.

(a) CIFAR-100 Undefended		(b) CIFAR-100 MemGuard	
Attack	Accuracy	Attack	Accuracy
Gap attack	83.5	Gap attack	83.5
Confidence-vector	88.1	Confidence-vector	50.0
Data augmentation	84.6	Data augmentation	84.6
Boundary distance	88.0	Boundary distance	88.0
Combined	89.2	Combined	89.2
(c) MNIST Undefended		(d) MNIST MemGuard	
Attack	Accuracy	Attack	Accuracy
Gap attack	53.2	Gap attack	53.2
Confidence-vector	55.7	Confidence-vector	50.0
Data augmentation	53.9	Data augmentation	53.9
Boundary distance	57.8	Boundary distance	57.8
Combined	58.7	Combined	58.7

2018a), fail to prevent label-only attacks, and thus, do not significantly reduce MI compared to an undefended model. Other proposed defenses, e.g., reducing the precision or cardinality of the confidence-vector (Shokri et al., 2016; Truex et al., 2018; Salem et al., 2018), and recent defenses like prediction purification (Yang et al., 2020), also rely on confidence masking: they are unlikely to resist label-only attacks. See Supplement § D for more details on these defenses.

5.3. Breaking MemGuard

We implement the strongest version of MemGuard that can make *arbitrary* changes to the confidence-vector while leaving the model’s predicted label unchanged. Observing Figure 1 and Table 1 (b) and (d), we see that MemGuard successfully defends against prior confidence-vector attacks, but as expected, offers no protection against our label-only attacks. All our attacks significantly outperform the (non-adaptive) confidence-vector and the baseline gap attack.

The main reason that Jia et al. (2019) found MemGuard to protect against confidence-vector attacks is because these attacks were not properly *adapted* to this defense. Specifically, MemGuard is evaluated against confidence-vector attacks that are tuned on source models *without MemGuard enabled*. This observation also holds for other defenses such as Yang et al. (2020). Thus, these attacks’ membership predictors are tuned to distinguish members from non-members based on high confidence scores, which these defenses obfuscate. In a sense, a label-only attack like ours is the “right” adaptive attack against these defenses: since the model’s confidence scores are no longer reliable, the adversary’s best strategy is to use hard labels, which these defenses explicitly do not

Table 2. Accuracy of membership inference attacks on Texas, Purchase, Location, and Adult. Where augmentations may not exist, noise robustness can still perform on or near par with confidence-vector attacks. The target models are trained exactly as in (Shokri et al., 2016): 1,600 points for Location and 10,000 for the rest. Our noise robustness attack uses 10,000 queries. Tables (a), (c), (e), and (g) report results without any defense. Tables (b), (d), (f), and (h) report results with MemGuard (Jia et al., 2019), which prevents the confidence-vector attacks via “confidence-masking”. Results that are affected by confidence masking are marked in red.

(a) Texas Undefended		(b) Texas MemGuard	
Attack	Accuracy	Attack	Accuracy
Gap attack	73.9	Gap attack	73.9
Confidence-vector	84.0	Confidence-vector	50.0
Noise Robustness	80.3	Noise Robustness	80.3

(c) Purchase Undefended		(d) Purchase MemGuard	
Attack	Accuracy	Attack	Accuracy
Gap attack	67.1	Gap attack	67.1
Confidence-vector	86.1	Confidence-vector	50.0
Noise Robustness	87.4	Noise Robustness	87.4

(e) Location Undefended		(f) Location MemGuard	
Attack	Accuracy	Attack	Accuracy
Gap attack	72.1	Gap attack	72.1
Confidence-vector	92.6	Confidence-vector	50.0
Noise robustness	89.2	Noise Robustness	89.2

(g) Adult Undefended		(h) Adult MemGuard	
Attack	Accuracy	Attack	Accuracy
Gap attack	58.7	Gap attack	58.7
Confidence-vector	59.9	Confidence-vector	50.0
Noise Robustness	58.7	Noise Robustness	58.7

modify. Moving forward, we recommend that the trivial gap baseline serve as an indicator of confidence masking: a confidence-vector attack should not perform significantly worse than the gap attack for a defense to protect against MI. Thus, to protect against (all) MI attacks, a defense cannot solely post-process the confidence-vector—the model will still be vulnerable to label-only attacks.

5.4. Breaking Adversarial Regularization

The work of Nasr et al. (2018a) differs from MemGuard and prediction purification in that it does not simply obfuscate confidence vectors at test time. Rather, it jointly trains a target model and a defensive confidence-vector MI classifier in a min-max fashion: the attack model to maximize MI and the target model to produce accurate outputs that yet fool

the attacker. See Supplement § D for more details.

We train a target model defended using adversarial regularization, exactly as in (Nasr et al., 2018a). By varying its hyper-parameters, we achieve a defended state where the confidence-vector attack is within 3 percentage points of chance, as shown in Supplement Figure 9. Again, our label-only attacks significantly outperform this attack (compare Figures 6 (a) and (b)) because the train-test gap is only marginally reduced; this defense is not entirely ineffective—it prevents label-only attacks from exploiting beyond 3 percentage points of the gap attack. However, when label-only attacks are sufficiently defended, it achieves significantly worse test accuracy trade-offs than other defenses (see Figure 5). And yet, evaluating the defense solely on confidence-vector attacks overestimates the achieved privacy.

5.5. The Query Complexity of Label-Only Attacks

We now answer question 3) and investigate how the query budget affects the success rate of different label-only attacks.

Recall that our rotation attack evaluates $N = 3$ queries of images rotated by r° and our translation attack $N = 4d + 1$ for shifts satisfying a total displacement of d . Figure 2 (a)-(b) shows that there is a range of perturbation magnitudes for which the attack exceeds the baseline (i.e., $1 \leq r \leq 8$ for rotations, and $1 \leq d \leq 2$ for translations). When the augmentations are too small or too large, the attack performs poorly because the augmentations have a similar effect on both train and test samples (i.e., small augmentations rarely change model predictions and large augmentations often cause misclassifications). An optimal parameter choice (r and d) outperforms the baseline by 3-4 percentage-points, which an adversary can tune using its local source model. As we will see in § 6, these attacks outperform all others on models that used data augmentation *at training time*.

In Figure 2 (c), we compare different boundary distance attacks, discussed in § 3.4. With $\approx 2,500$ queries, the label-only attack matches the white-box upper-bound using $\approx 2,000$ queries and also matches the best confidence-vector attack (see Figure 1). With $\approx 12,500$ queries, our combined attack can outperform all others. Query limiting would likely not be a suitable defense, as Sybil attacks (Douceur, 2002) can circumvent it; even in low query regimes (< 100) our attacks outperform the trivial gap by 4 percentage points. Finally, with < 300 queries, our simple noise robustness attack outperforms our other label-only attacks. At large query budgets, our boundary distance attack produces more precise distance estimates and outperforms it. Note that the monetary costs are modest at $\approx \$0.25$ per sample⁵.

⁵<https://www.clarifai.com/pricing>

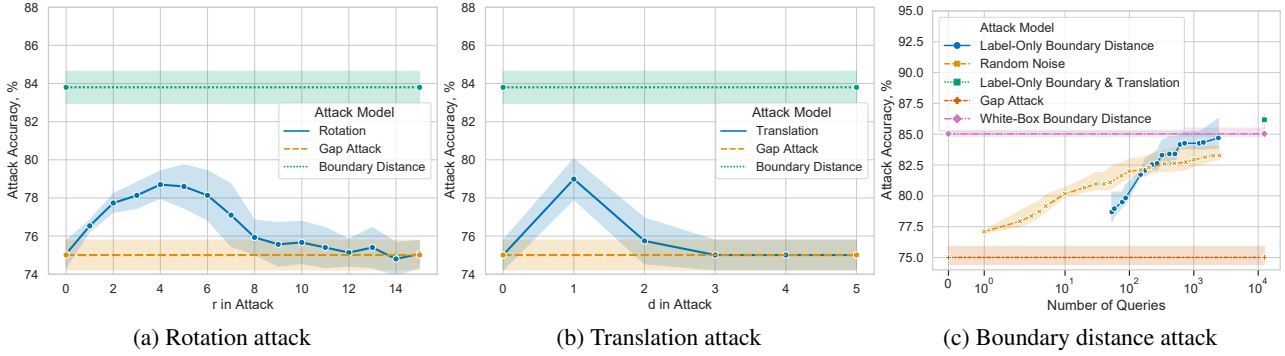


Figure 2. **Comparison of query settings for different label-only MI attacks on CIFAR-10.** Target model are trained on a subset of 2,500 data points. In (a) and (b), we compare the performance of the data augmentation attack against two baselines (the gap attack and the label-only boundary distance attack, with 2,500 queries), as we increase the r and d parameters. In (c), we compare attacks that threshold on a point’s distance to the boundary in a black-box setting with a white-box baseline using Carlini and Wagner’s attack (Carlini & Wagner, 2017). We describe these attacks in § 3.4. Costs are $\approx$ \$0.1 per 1000 queries⁵.

6. Defending with Better Generalization

Since confidence-masking defenses cannot robustly defend against MI attacks, we now investigate to what extent standard regularization techniques—that aim to limit a models’ ability to overfit to its training set—can. We study how data augmentation, transfer learning, dropout (Srivastava et al., 2014), ℓ_1/ℓ_2 regularization, and differentially private (DP) training (Abadi et al., 2016) impact MI.

We explore three questions in this section:

- How does training with data augmentation impact MI attacks, especially those that evaluate augmented data?
- How well do other standard machine learning regularization techniques help in reducing MI?
- How do these defenses compare to differential privacy, which can provide formal guarantees against any form of membership leakage?

6.1. Training with Data Augmentation Exacerbates MI

Data augmentation is commonly used in machine learning to reduce overfitting and encourage generalization, especially in low data regimes (Shorten & Khoshgoftaar, 2019; Mikołajczyk & Grochowski, 2018). Data augmentation is an interesting case study for our attacks. As it reduces a model’s overfitting, one would expect it to reduce MI. But, a model trained with augmentation will have been trained to strongly recognize x and its augmentations, which is precisely the signal that our data augmentation attacks exploit.

We train target models with data augmentation similar to § 3.3 and focus on translations as they are most common in computer vision. We use a simple pipeline where all translations of each image is evaluated in a training epoch. Though this differs slightly from the standard random sampling, we choose it to illustrate the maximum MI when the adversary’s

queries exactly match the samples seen in training.

Observe from Figure 3 that augmentation reduces overfitting and improves generalization: test accuracy increases from 49.7% without translations to 58.7% at $d = 5$ and the train-test gap decreases. Due to improved generalization, the confidence-vector and boundary distance attacks’ accuracies *decrease*. Yet, the success rate of the data augmentation attack *increases*. This increase confirms our initial intuition that the model now leaks *additional* membership information via its invariance to training-time augmentation. Though the model trained with $d = 5$ pixel shifts has higher test accuracy, our data augmentation attack exceeds the confidence-vector performance on the non-augmented model.⁶ Thus, model generalization is not the only variable affecting its membership leakage: models that overfit less on the original data may actually be more vulnerable to MI *because they implicitly overfit more on a related dataset*.

Attacking a high-accuracy ResNet We use, without modification, the pipeline from FixMatch (Sohn et al., 2020), which trains a ResNet-28 to 96% accuracy on the CIFAR-10 dataset, comparable to the state of the art. As with our other experiments, this model is trained using a subset of CIFAR-10, which sometimes leads to observably overfit models indicated by a higher gap attack accuracy. We train models using four regularizations, all random: vertical flips, shifts by up to $d = 4$ pixels, image cutout (DeVries & Taylor, 2017), and (non-random) weight decay of magnitude 0.0005. All are either enabled or disabled.

Our results here, shown in Figure 4 corroborate those ob-

⁶Though we find in Supplement Figure 8 that the attack is strongest when the adversary correctly guesses d , we note that these values are often fixed for a domain and image resolution. Thus, adversarial knowledge of the augmentation pipeline is not a strong assumption.

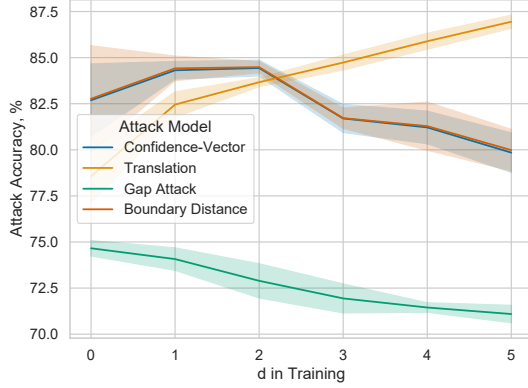


Figure 3. Accuracy of MI attacks on CIFAR-10 models trained with data augmentation on a subset of 2500 images. As in our attack, d controls the number of pixels by which images are translated during training, where no augmentation is $d = 0$. For models trained with significant amounts of data augmentation, MI attacks become *stronger* despite it generalizing better.

tained with the simpler pipeline above: *though test accuracy improves, our data augmentation attacks match or outperform the confidence-vector attack.*

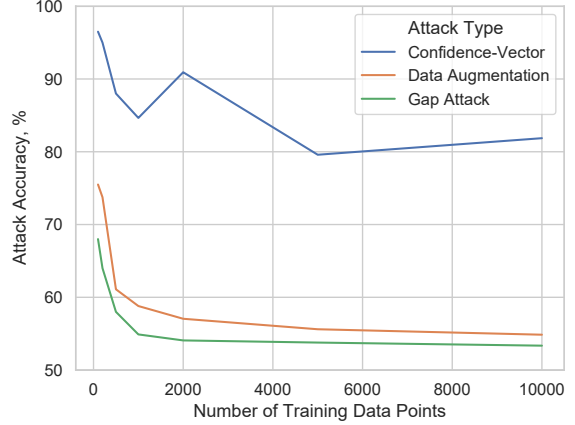
6.2. Other Techniques to Prevent Overfitting

We explore questions B)-C) using other standard regularization techniques, with details in Supplement E. In transfer learning, we either only train a new last layer (*last layer fine-tuning*), or fine tune the entire model (*full fine-tuning*).

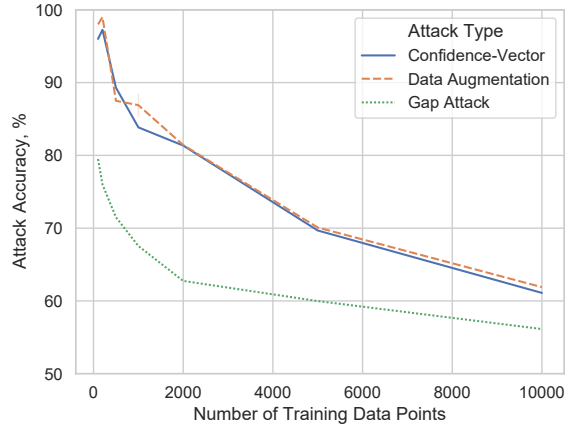
We pre-train a model on CIFAR-100 to 51.6% test accuracy and then use transfer learning. We find that boundary distance attack performed on par with the confidence-vector in all cases. We observe that last layer fine-tuning degrades all our attacks to the generalization gap, preventing non-trivial MI (see Figure 10 in Supplement § F). This result corroborates intuition: linear layers have less capacity to overfit compared to neural networks. We observe that full fine-tuning leaks more membership inference but achieves better test accuracies, as shown in Figure 5.

Finally, DP training (Abadi et al., 2016) formally enforces that the trained model does not strongly depend on any individual training point—that it does not overfit. We use differentially private gradient descent (DP-SGD) (Abadi et al., 2016) (see Supplement § E). To achieve comparable test accuracies as undefended models, the formal privacy guarantees become mostly meaningless (i.e., $\epsilon > 100$).

In Figure 6, we find that most forms of regularization fail to prevent even the baseline gap attack from reaching 60% accuracy or more. Only strong ℓ_2 regularization ($\lambda \geq 1$) and DP training consistently reduced MI. Figure 5 gives us a best understanding of the privacy-utility trade-off. We see that both prevent MI at a high cost in test-accuracy—



(a) Without Augmentations



(b) With Augmentations

Figure 4. Accuracy of membership inference attacks on CIFAR-10 models trained as in FixMatch (Sohn et al., 2020). Our data augmentation attacks, which mimic the training augmentations, match or outperform the confidence-vector attacks when augmentations were used in training. We evaluate 1000 randomly generated augmentations for this attack. As in previous experiments, we find our label-only distance attack performs on par with the confidence-vector attack. Interestingly, the gap attack accuracy also improves due to a relatively larger increase in training accuracy. “With Augmentations” and “Without Augmentations” refer to using all regularizations, as in FixMatch, or none, respectively.

they cause the model to *underfit*. However, we also clearly see the utility benefits of transfer learning: these models achieve consistently better test-accuracy due to features learned from non-private data. Combining DP training with transfer learning mitigates privacy leakage at only minimal cost in test accuracy, achieving the best tradeoff. When transfer learning is not an option, dropout performs better.

7. Worst-Case (Outlier) MI

Here, we perform MI only for a small subset of “outliers”. Even if a model generalizes well on average, it might still

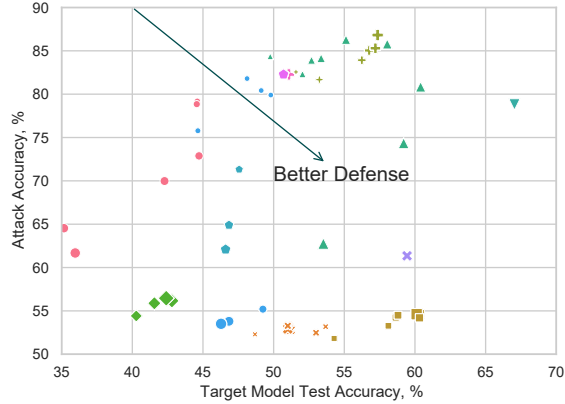


Figure 5. Test accuracy and label-only attack accuracy for various defenses. Same setup as Figure 6. Models towards the bottom right are more private and more accurate.

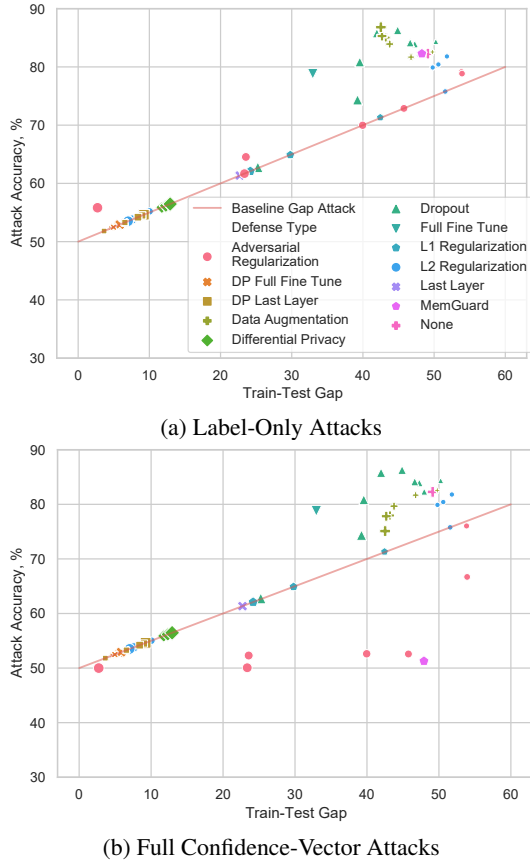


Figure 6. Comparison of MI attacks on various defenses. Target models are trained on 2500 data points from CIFAR-10. Point sizes indicate relative regularization amounts within a defense.

have overfit to unusual data in the tails of the distribution (Carlini et al., 2019). We use a similar but modified process as Long et al. (2018) to identify potential outliers.

First, the adversary uses a source model $\hat{h}$ to map each targeted data point, x , to its feature space, or the activations of its penultimate layer, denoted as $z(x)$. We define two

points x_1, x_2 as *neighbors* if their features are close, i.e., $d(z(x_1), z(x_2)) \leq \delta$, where $d(\cdot, \cdot)$ is the cosine distance and δ is a tunable parameter. An outlier is a point with less than γ neighbors in $z(x)$ where γ is another tunable parameter. Given a dataset X of potential targets and an intended fraction of outliers β (e.g., 1% of X), we tune δ and γ so that a β -fraction of points $x \in X$ are outliers. We use precision as the MI success metric.

We run our attacks on the outliers of the same models as in Figure 6. We find in See Figure 11 in Supplement Section F, that we can always improve the attack by targeting outliers, but that strong $L2$ regularization and DP training prevent MI. As before, we find that the label-only boundary distance attack matches the confidence-vector attack performance.

8. Conclusion

We developed three new label-only membership inference attacks that can match, and even exceed, the success of prior confidence-vector attacks, despite operating in a more restrictive adversarial model. Their label-only nature requires fundamentally different attack strategies, that—in turn—cannot be trivially prevented by obfuscating a model’s confidence scores. We have used these attacks to break two state-of-the-art defenses to membership inference attacks.

We have found that the problem with these “confidence-masking” defenses runs deeper: they cannot prevent any label-only attack. As a result, any defenses against MI necessarily have to help reduce a model’s train-test gap.

Finally, via a rigorous evaluation across many proposed defenses to MI, we have shown that differential privacy (with transfer learning) provides the strongest defense, both in an average-case and worst-case sense, but that it may come at a cost in the model’s test accuracy.

To center our analysis on comparing the confidence-vector and label-only settings, we use the same threat model as prior work (Shokri et al., 2016) and leave a fine-grained analysis of label-only attacks under reduced adversarial knowledge (e.g., reduced data and model architecture knowledge (Yeom et al., 2018; Salem et al., 2018)) to future work.

Acknowledgments

We thank the reviewers for their insightful feedback. This work was supported by CIFAR (through a Canada CIFAR AI Chair), by NSERC (under the Discovery Program, NFRF Exploration program, and COHESA strategic research network), and by gifts from Intel and Microsoft. We also thank the Vector Institute’s sponsors.

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